

Generative Reward Synthesis

for Open-Ended Navigation with Autonomous Systems

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Motivation & Thesis Goal

The Problem

- ① RL requires hand-engineered reward functions — doesn't scale to open-ended language instructions
- ② LLMs have rich semantic knowledge but no direct path to low-level robot control
- ③ Letting LLMs write control code directly is often slow and unsafe

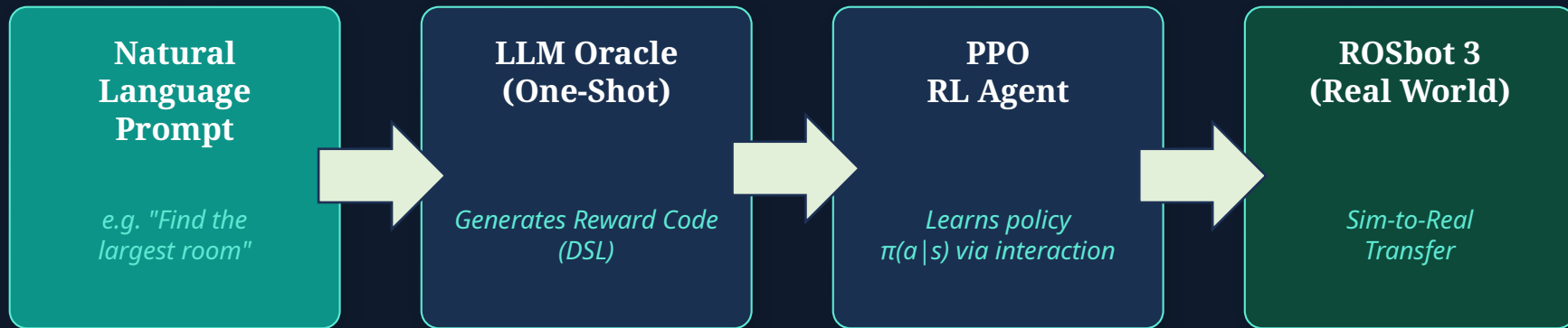
Our Approach

Generative Reward Synthesis

LLM as a one-shot compiler:

- > Reads a natural-language task
- > Generates executable reward code (DSL)
- > RL agent trains fully autonomously
- > Policy deploys on real ROSbot 3

System Architecture



Key Principle:

The LLM is a compiler, not a runtime participant. Once Reward function is handed off, PPO trains fully autonomously .

Current Status

Completed

Actor-Critic Foundation

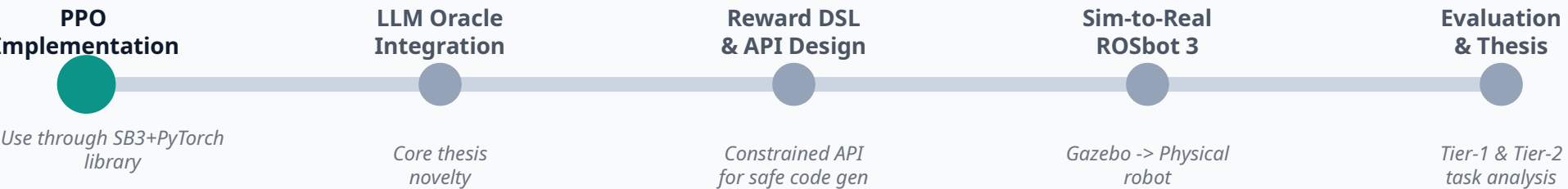
- ✓ MDP formally defined
— state, action, reward, discount
- ✓ Actor-Critic architecture implemented end-to-end
- ✓ Core RL loop: observe, act, receive reward, update
- ✓ TD-learning and value estimation working in code

In Progress

PPO Implementation

- > Extending Actor-Critic to full PPO with clipped surrogate objective
- > Generalised Advantage Estimation (GAE) integrated
- > Shared-backbone neural network (actor + critic heads)
- > Goal: validate PPO in simulation before integrating LLM Oracle

Roadmap & Plan



Open Research Questions

Q1 Reward DSL Granularity

How expressive must the constrained API be so the LLM can generate reward functions that are both safe to execute and rich enough to train a policy? What sub-reward primitives are necessary and sufficient for navigation tasks?

Q2 Sim-to-Real Transfer

Can a policy trained entirely in Gazebo simulation transfer to a physical ROSbot 3 without additional fine-tuning? What are the biggest sources of the sim-to-real gap for this robot platform?

Q3 Evaluation & Success Metrics

How should success be defined and measured across evaluation tiers? Tier-1 (object search) and Tier-2 (topological navigation) may require different metrics — what thresholds distinguish meaningful task completion?

Summary

Goal

LLM generates reward code -> PPO learns autonomously -> Deployment on ROSbot 3

Architecture

LLM as one-shot compiler + PPO as the learning system

Status

Actor-Critic complete. PPO integration underway .

Next Steps

PPO (via SB3 + PyTorch) -> LLM Oracle & DSL design -> Reward synthesis pipeline -> Sim-to-Real

Thank you — Questions?